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Electricity Theft Detection for Smart Homes: Harnessing the Power of Machine Learning with Real and Synthetic Attacks

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ABSTRACT A new framework for detecting electricity theft in smart homes using machine learning has been proposed. Traditionally, theft detection relied on manual inspections and meter failures, but with the advancement of machine learning, it is now possible to automatically detect theft based on meter reading patterns. However, the accuracy of detection is a concern due to the various factors that influence electric consumption, such as lifestyle and weather. The proposed framework utilizes knowledge-based synthetic attack data to train an attack classifier. This data consists of benign and attack patterns, which serve as the foundation for generating synthetic and simulated attacks, closely mirroring real-world scenarios. The framework was evaluated using the Almanac of Minutely Power dataset version 2 (AMPds2), which contains fine-grained time-series data from a smart home. We preprocessed this data for binary classification to detect anomalies to validate our synthetic attack approach with a real attack dataset. The Extreme Gradient Boosting binary-based discriminator algorithm performs best with average 98.74% and 98.69% AUC scores for real and simulated attacks for all homes in detecting and classifying anomalies while Random Forest performed next with average AUC scores of 97.07% and 96.76% for the same attack scenario respectively. These methods outperformed legacy unsupervised approaches.

INDEX TERMS Electricity theft, Machine learning, Synthetic attack data, Real attack data, Detection models, Unsupervised learning, Supervised learning.

I. INTRODUCTION

ELECTRICITY theft is a pervasive issue with economic implications, necessitating innovative approaches for its detection; although invisible, possesses substantial economic value and is an essential resource for modern society. The issue of electricity theft, often referred to as Electricity Theft Attacks (ETAs), represents a pervasive and costly problem worldwide, particularly in developing countries [1], [2]. In India, for instance, it is estimated that more than a fifth of the total electricity production is lost due to theft [3]. Moreover, within the United States, these losses are estimated at approximately $6 billion annually. [4], [5]. The detection of ETAs traditionally relies on human intervention through routine inspections of power meters and cables, user-reported suspicions, or identification of meter device failures [6]. Nevertheless, these methods are human-centric and struggle to effectively pinpoint instances of electricity theft, primarily due to the inherent complexity of power systems.

Each electrical appliance exhibits a distinctive consumption signature in the time series data, which is aggregated with the consumption patterns of other appliances at the metering point [7]. Consequently, the output of a smart meter encapsulates the collective patterns of electricity consumed by legitimate and potentially malicious loads [8]. This aggregation poses a challenge for the detection of electricity theft through traditional heuristic analysis of meter readings alone [9]. However, with the rapid advancement of machine learning technologies, there is a burgeoning opportunity to harness the power of artificial intelligence in detecting electricity theft from these aggregated consumption patterns.

This paper introduces an innovative Electricity Theft Detection (ETD) framework tailored for smart homes, representing a notable departure from traditional approaches by integrating knowledge-based synthetic attack data. Our approach is particularly well-suited for handling unlabeled datasets, which are commonly addressed using unsupervised learning techniques [10]. While synthetic data generation is often considered a form of supervised learning due to the creation of labels through synthetic approach [11], [12], it shares applicability with unsupervised methods when dealing with unlabeled data. However, our research distinctly demonstrates the superior performance of our approach compared to conventional unsupervised learning techniques. This crucial finding underscores the primary message conveyed by this paper.

One of the key challenges in deploying machine learning classifiers for ETD is the pervasive issue of data imbalance [13], [14], where the volume of normal and abnormal samples exhibits substantial disparities. While benign samples are readily accessible through historical data, attack or theft samples are often scarce, and in some cases, entirely absent for certain customers.

In response to this dilemma, we address the issue of imbalanced data and zero-day attacks [15] through the creation of a synthetic attack dataset. Leveraging the inherent predictability of theft patterns (Jokar et al., 2015) [5], this innovative approach significantly elevates the detection rate and empowers the identification of a diverse range of attack types.

The prospect of training an ETD machine learning model using synthetic attack data offers a range of advantages. Firstly, it circumvents the need for extensive time and resource investments, which would otherwise be required to acquire authentic attack consumption patterns alongside legitimate patterns in real-life scenarios. This cost-effective approach also allows for the evaluation of synthetic attack scenarios with real data of a similar nature. Secondly, the calculation of attack-contained meter readings becomes straightforward by simply adding synthetic malicious consumption to legitimate usage. Such a practical application was previously unattainable in computer network communications, where node behavior hinged on the presence of attacks. Additionally, the flexibility of synthetic attack data, drawing from a broad spectrum of knowledge, holds the potential to significantly enhance the training of robust ETD models for precise attack classifications.

While existing literature (Punmiya et al., 2019) [16] has predominantly relied on simple statistical models such as random, mean, shift, and reverse for synthetic attack loads; our approach, an extension of our prior research, wherein we introduce a novel set of attack scenarios such as Baseload, Weakload, Peakhour, Midnight, and Evil-Twin attacks, offers a more realistic framework, rooted in the intentions behind various types of attacks as [17]. We validated with real attack data by prepossessing our synthetic data for binary classification anomaly detection.

It is essential to distinguish between the protection of utility companies (Dayaratne et al., 2021 [18]; Singh et al., 2017 [19]; Punmiya et al., 2019 [16]) and safeguarding individual homes, as both endeavors aim to combat electricity theft and power stealing attacks. When an attacker directly steals power from a utility company by bypassing or tampering with the meter, it results in a decrease in the home's electricity bills. However, the origin of the stolen electricity may not necessarily be from the grid; the attacker might steal power from another home, leading to an increase in their electricity bills. In such instances, households must autonomously detect and protect against theft, as utility companies might not directly address the consumption increases resulting from electricity theft scenarios.

In our setup, we extracted raw historical electricity appliance consumption data of a residential house in Canada from 2012 to 2014 [20] and injected a binary simulated attack. We trained our synthetic (SYN-XGB), (SYN-RF), (SYN-MLP), classifiers models and unsupervised (UN-MLP-AE), (UN-1D-CONV-AE) and (UN-IF) models for different homes with benign and attack classes. The proposed model, SYN-XGB, effectively classifies benign and attack instances with high average detection AUC scores 98.69% for simulated, 98.73% for real attacks respectively, and with low false positive and false negative detection rates.

In our validated binary classification, it is noteworthy that all binary classifiers excel in predicting anomalies, with SYN-XGB yielding the most favorable results when compared to unsupervised autoencoders (UN-MLP-AE and UN-1D-CNN-AE) anomalies detection and Isolation Forest (UN-IF) for outlier detection.

Our contributions to this paper are closely tied to the preprocessed extraction of electricity appliances consumption pattern from the AMPds2 dataset [20], which enabled the generation of synthetic attacks, simulated attacks, and the inclusion of real attacks in the datasets:

- **Enhanced Dataset Preprocessing for Synthetic Attacks:** Our paper greatly improves AMPds2 dataset preprocessing by introducing a systematic method to generate synthetic attack scenarios. This involves adding synthetic attacks to the benign electricity consumption data, enriching the dataset, and enabling more robust supervised training for anomaly detection models.
- **Real-World evaluation through Real Attacks:** The addition of real attack data with synchronized time features to the synthetic attacks is a unique and critical contribution. It provides real-world evaluation for the synthetic attack scenarios, ensuring the effectiveness of the detection models in distinguishing simulated attacks from actual anomalies in electricity consumption patterns.
- **Advancements in Smart Grid Cybersecurity:** The improved dataset preprocessing, combining synthetic, simulated, and real attack data, enhances the development

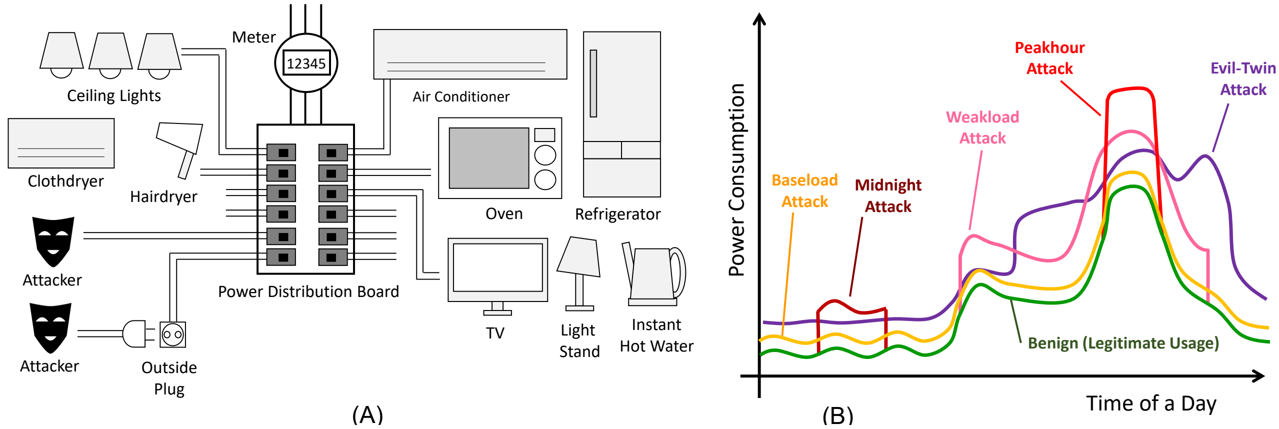


FIGURE 1: A: Power Distribution Board with connected sample appliances. B: Scenarios of electricity theft attacks. Power distribution boards and cables are deployed behind walls and are usually not visible. An attacker uses the pre-deployed cable in a complex building or the outside plug to steal electricity. Depending on the theft pattern, we identify five classes of attacks.

of robust detection models. This empowers smart homeowners, utility companies, and researchers to enhance smart grid security against electricity theft and other threats.

The rest of the paper is organized as follows. Section II discusses related works. Section III describes the attack model before proceeding to Section IV to explain our proposal: ETD with Synthetic Attack Data(SAD). Section V contains ETD Experiments setting. Section VI contains the experiment results and performance evaluation. In Section VII, we provide discussions and future works, and Section VIII concludes this paper.

II. RELATED WORKS

Aldegheishem et al. [21] have identified five distinct methods of electricity theft attacks from the perspective of utility companies: (1) meter bypass, (2) meter hacking, (3) direct hooking, (4) tapping, and (5) meter tampering. Additionally, the concept of a False Data Injection Attack (FDIA) for electricity theft [18], [19] has been discussed as a cyber-attack within the realm of cyber-physical systems. Jokar et al. [5] have also included FDIA alongside physical attacks that involve bypassing or tampering with meters, all of which manipulate meter readings to reduce electricity consumption [22]. Consequently, the focus of attention has shifted towards electricity theft detection through the analysis of meter readings.

Various machine-learning techniques have been explored for classifying electricity consumption patterns and detecting electricity theft. These methods include Support Vector Machines (SVM) [5], [23], Decision Trees [24], and more recently, Gradient-Boosting [16], [25]. These approaches typically adopt supervised learning principles, relying on labeled datasets containing both benign and attack data samples. However, obtaining ground truth labels for real-life scenarios can be a formidable challenge.

An alternative approach for electricity theft detection, particularly when labeled data is unavailable, is anomaly detection. This method encompasses anomaly pattern detection based on hypothesis testing (APD-HT) [26], hierarchical self-organizing maps (SOM) [27], and stacked sparse denoising autoencoders [28]. Anomaly detection, while effective, has the drawback of flagging any abnormal occurrences, including instances like an unusual appliance usage pattern during nighttime [29].

Our work operates within the context of real-life scenarios where labeled data is scarce. However, we introduced a novel approach by incorporating knowledge of potential attack scenarios and synthetic attack data to train a supervised model using a non-labeled real-world dataset. Furthermore, our focus is on securing homes from the threat of electricity theft by external parties. In cases where an attacker steals electricity not directly from the power grid but from neighboring houses, the utility company may remain indifferent to such theft, leaving homeowners responsible for safeguarding their own resources.

It's worth noting that the work of Punmiya et al. [16] shares similarities with our approach, as they also propose Gradient Boosting and synthetic theft patterns. However, their primary aim is to protect the interests of the utility company, where theft patterns are designed to reduce electricity bills. In contrast, our scenario involves theft patterns that increase the bills for homeowners. The existing literature predominantly considers methods such as random-based reduction, mean-based reduction, and reverse-time-based reduction for training binary classification models between benign and attack data, as established in prior work [5]. In contrast, our paper endeavors to simulate more realistic electricity theft patterns and validate trained synthetic binary classification models to predict anomalies or outliers with real attack data, if detected, which represents a novel and more complex approach. Furthermore, our work capitalizes on fine-grained time-series data within a smart home environment, a resource not currently available in today's smart grid landscape.

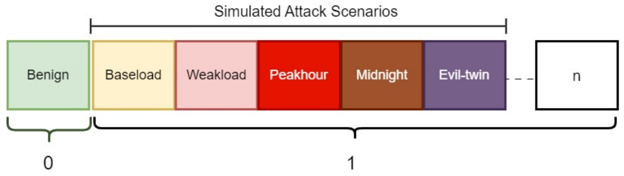


FIGURE 2: Schematic for anomaly detection.

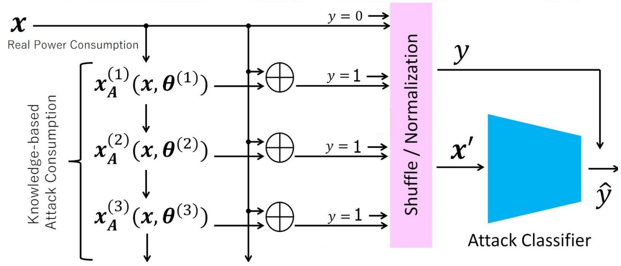


FIGURE 3: The framework for electricity theft detection with synthetic attack data. Without real attack data, possible attack patterns can be theoretically generated and added onto real power consumptions for training an attack classifier.

III. ELECTRICITY THEFT ATTACK

This paper is an extension of our previous study, Abraham et al [17], where we presented a comprehensive framework for ETD in smart homes using knowledge-based synthetic attack data(KBSAD). The study was primarily focused on the multiclass classification of attack scenarios using fine-grained time-series electricity consumption appliances with data points as shown in Table 1 from AMPds2 dataset [20]. The five attack types, as shown in Figure 1B and visualized in Figure 4, simulate various forms of electricity theft generated and their corresponding attack impacts as seen in Table 2 before training data augmentation.

1) **Baseload Attack:** This type of attack involved a consistent increase in electricity consumption, mimicking a situation where an attacker maintains a constant load over time. We introduced a 100-unit increase in power consumption during the attack period.
2) **Weakload Attack:** For this attack, we targeted situations where an attacker steals off a fraction of the electricity consumption. If the baseline consumption exceeded 500 units, we introduced a 500-unit consumption reduction to simulate the theft.
3) **Peakhour Attack:** Peak-hour attacks mimic scenarios where attackers steal electricity during high-demand periods. We introduced a sudden 2,000-unit increase in consumption if the baseline exceeded 1,500 units during specific hours.
4) **Midnight Attack:** Midnight attacks represent theft during late-night hours. We randomly selected a 2-hour window and added a 1,000-unit consumption increase during this time.
5) **Evil-Twin Attack:** This sophisticated attack type introduced the concept of mimicking the consumption pattern of another home, the "evil twin." Consumption data from another home was overlaid onto the baseline data to simulate this scenario.

The inclusion of real test data further solidifies the evaluation of our model's performance.

In this section, we extend our research by transitioning from multiclass to binary classification, incorporating data from three distinct homes (Home A, Home B, and Home C as described in section IV), and validating our models with a real attack dataset.

A. ATTACK SCENARIO

A house has an electric meter for accounting purposes. The household has to pay the bill based on the accumulated power counted by the meter. An attacker may physically access the power distribution board under the electric meter and steal power from the house to make the household pay the bills on their behalf. An attacker may also steal power from an outlet if it is physically accessible outside as depicted in Figure 1A.

This kind of scenario may happen in apartments, congested complex buildings, and/or office rooms where power line cables are not easily traceable. For example, some houses might be leased with pre-deployed lines for electricity theft.

In this paper, we have preprocessed the five attack classes in our previous work, that is, 'baseload', 'weakload', 'peakhour', 'mid-night', and 'evil twins' and other unclassified attacks, 'n' as '1' as shown in Figure 2. The items mentioned above are regarded as crucial because they have the potential to affect to a great extent the billings of the attacked household consistently if the attacker is not discovered. We regenerated two datasets from the baseline dataset and a real test dataset for each home scenario for comparison, one in which we collected data from the household consumption pattern(benign) without any attack; we obtained the second dataset (simulated test) and the third dataset (real test) by injecting attacks. This process was repeated for each of the homes. Since each home has a different appliance configuration as described in the next section, they have different attack instances. For example in Table 3, the attack instances generated for Home A was 33818 with the benign elements of 35040 consumption pattern. We injected real test data with uniform consumption electricity patterns [30] for true evaluation of our simulated attacks.

IV. ELECTRICITY THEFT DETECTION WITH SYNTHETIC ATTACK DATA

In our quest to detect electricity theft attacks effectively, we adopt a comprehensive approach that involves closely monitoring power consumption patterns. We emphasize the importance of granularity in both the time and power domains, particularly at the power aggregation point. In this research paper, we delve into the realm of synthetic and real attack data, leveraging their combined potential to validate machine learning applications. Our primary objective is to accurately identify genuine anomalies originating from legit-

TABLE 1: Appliances and units in AMPds

Point ID	Description	Point ID	Description
WHE	Whole House Meter	FRE	HVAC/Furnace
B1E	North Bedroom	GRE	Garage
B2E	Master/South BR	HPE	Heat Pump
BME	Basement Plugs/Lights	HTE	Instant Hot Water Unit
CDE	Clothes Dryer	OFE	Home Office
CWE	Clothes Washer	OUE	Outside Plug
DNE	Dinning Room Plugs	TVE	Entertainment TV/PVR/AMP
DWE	Dishwasher	UTE	Utility Room Plug
EBE	Electronics Workbench	WOE	Wall Oven
EQE	Security/Network	UNE	Unmetered Loads
FGE	Kitchen Fridge		.

imate electricity consumption patterns. Figure 3 shows the framework of our synthetic attack data learning for electricity theft detection.

Let us consider $\mathbf{x}$ – a vector of power consumption. This vector contains the power consumption of each timeslot in the time order. For example, $\mathbf{x}$ may represent a power consumption of a certain day, and the i-th element x_i corresponds to the power usage at i-th minute from the beginning of the day. In this case, $\mathbf{x}$ has 1440 elements: i.e., $60 \times 24 = 1440$.

In supervised learning, we assume that each $\mathbf{x}$ has a corresponding label y for training a classification model for power consumption patterns. In our case, we are considering unsupervised learning, thereby we can assume that label $y = 0$ as a benign case, and all other attacks such as Baseload attack, weakload attack, and so on are all labeled $y = 1$ as an attack case. In the real, practical scenario, we will only get a collection of $\mathbf{x}$ from a house as a result of long-term monitoring, and we will not get real attack-enabled cases as anomalies. However, many power-stealing cases can be simulated just by arithmetically adding stolen power as a power consumption.

Let $\mathbf{x}_A$ be a vector of stolen power by an attacker. As we assume the attacker changes the stealing power based on the consumption of the house, $\mathbf{x}_A$ is a function of $\mathbf{x}$ and attacking parameters θ: e.g., $\mathbf{x}_A(\mathbf{x}, \theta)$.

In the case of attack, i.e., $y = 1$, we consider to inject an attack randomly from the attack type z. Then, the stolen attack vector is: $\mathbf{x}_A^{(z)}(\mathbf{x}, \theta^{(z)})$.

Finally, we can get the binary classification (benign and attack) dataset as follows.

$$(\mathbf{x}', y) = \begin{cases} (\mathbf{x}, 0) & y = 0 \\ (\mathbf{x} + \mathbf{x}_A^{(z)}(\mathbf{x}, \theta^{(z)}), 1) & y = 1 \end{cases}. \tag{1}$$

Machine learning models can be applied to the collection of $(\mathbf{x}', y)$ for binary classification and outlier detection.

A. SYNTHETIC DATASET OVERVIEW

For our study, we utilized the AMPds2 dataset (Almanac of Minutely Power Dataset version 2) [20] as a benchmark, representing two years' worth of home power consumption data. AMPds2 includes minute-by-minute power measurements recorded at the outputs of power distribution boards as shown in Figure 1A. The monitoring points within this dataset are detailed in Table 1.

Given the varying configurations of homes, it's possible that certain appliances, such as Clothes Dryers, Wall Ovens, or Dishwashers, may not be present in some households. To address this variability, we simulated three distinct home types by excluding the power consumption of these optional appliances. These home types are as follows:

- **Home A** is composed of a full set of appliances, which is the aggregated power consumptions of B1E, B2E, BME, CWE, DNE, HTE, EBE, EQE, FRE, OFE, OUE, TVE, UTE, CDE, HPE, DWE, FGE, and WOE. Some of those appliances such as those associated with HPE and WOE make peak power consumption, which may allow the peak-hour attacker to steal power more efficiently.
- **Home B** excludes CDE and WOE from Home A, assuming that the existence of a cloth dryer and wall oven may influence the accuracy of attacker detection.
- **Home C** excludes DWE, HPE, and TVE from Home A, assuming that the existence of a dishwasher, heat pump, and small appliances may influence the accuracy of attacker detection.

Recognizing that electric power consumption data is inherently time-dependent, we adopted a data segmentation strategy. Specifically, we selected the initial 80% of the data, equivalent to 584 days, for our training dataset. During this phase, we applied our synthetic attack data methodology to enrich the dataset. The remaining 20% of the data, spanning 146 days, was reserved for our test dataset in our benchmark experiment as shown in Table 2. Importantly, the test dataset included simulated attacks generated as part of this study.

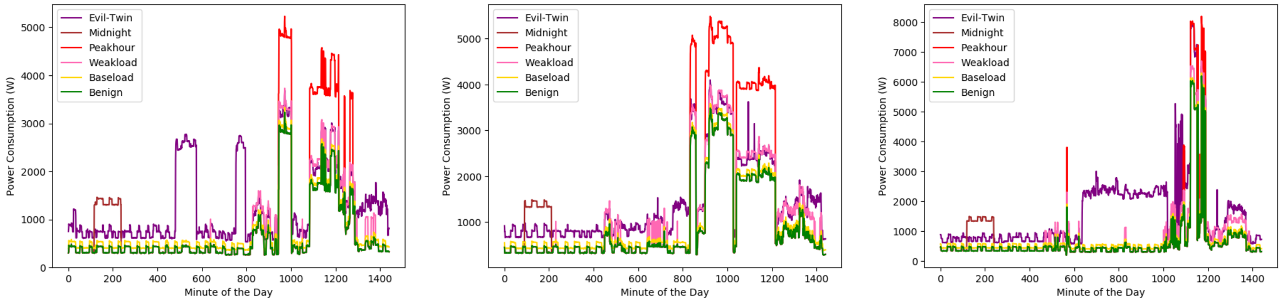
It's important to highlight that the test dataset does not cover an entire year. This deliberate choice was made to allocate a larger volume of data for the training phase, enhancing the robustness of our models.

B. PREPROCESSING FOR BINARY CLASSIFICATION

The electricity theft attack detection dataset (ETA-DD) consists of:

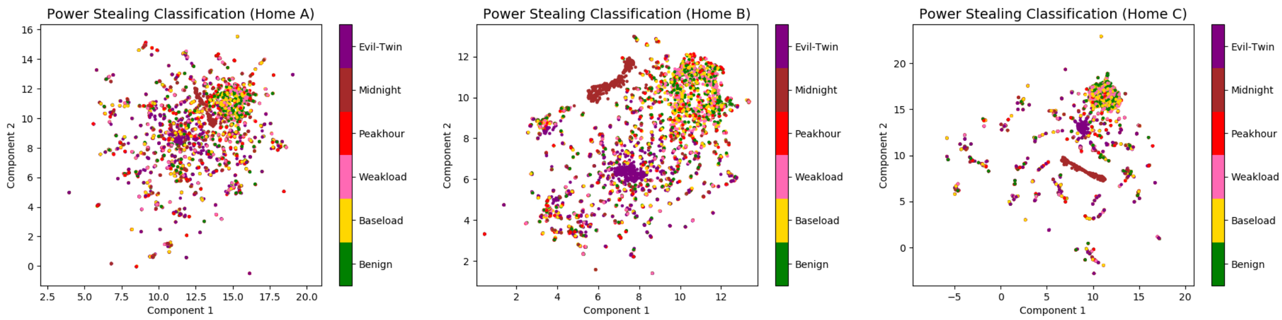
1) two training sub-datasets and
2) two testing sub-datasets for homes A, B & C

This ETA-DD is assumed for Binary Classification Problems (Benign or Attack). This is because it is intended for the application of unsupervised learning for attack detection, and the evaluation of detection accuracies with the real building data. The real building power consumption cannot be easily labeled for the predefined attack cases (in section III). These are the reasons why this study focuses on attack detection, rather than classification. Even though it is not intended for attack classification, the evaluation scope has drastically widened.



(a) Day 1 (b) Day 2 (c) Day 3

FIGURE 4: Electricity usage of Home A with synthetic attack data on different days. The benign consumption patterns drastically change day by day. Especially, The Weakload and Peakhour attacks are sophisticated – having similar consumption patterns with the legitimate usage although the values are different.



(a) Home A (b) Home B (c) Home C

FIGURE 5: 2D projections of attack contained electricity usage with the benign case by uniform manifold approximation and projection (UMAP) [31]. Each plot corresponds to the consumption pattern of a day on different attack scenarios. Depending on the available appliances of the homes, the shapes of the attack classes are very different. There are more overlaps among classes in Home A, whereas we observe fewer overlaps in Home C.

TABLE 2: The Profile of the dataset generated with synthetic attack data, and corresponding attack impacts (AI)

Home	Category	Benign	Baseload	Weakload	Peakhour	Midnight	Evil-Twin	Averaged AI	AI Ratio
A	Train	584	584	580	490	584	584	1.15 USD	1.77
	Test	146	146	146	140	146	146	1.17 USD	1.72
B	Train	584	584	580	437	584	584	1.08 USD	1.80
	Test	146	146	146	130	146	146	1.10 USD	1.77
C	Train	584	584	576	306	584	584	0.78 USD	1.66
	Test	146	146	146	84	146	146	0.76 USD	1.68

1) Feature Vector (X) and Label (Y)

Each file starts from a header row, and then consumption records follow. Each consumption record is organized as Figure 6.

X: features consist of 1380 elements each corresponding to the power consumption of the minute of the day. However, 1380 minutes are 23 hours. This is because we have applied data augmentation as described in Data Augmentation subsection below.

Y: a label that identifies the meaning of the X. Benign is defined as 0. Attack is defined as 1.

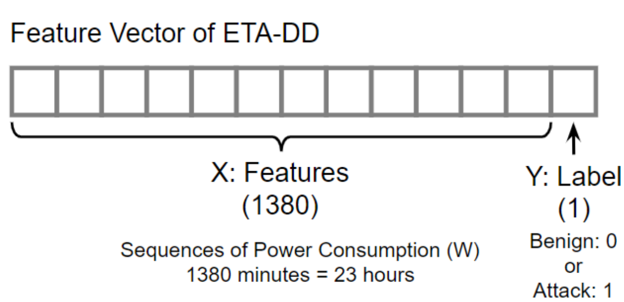


FIGURE 6: Features Vector and Labels

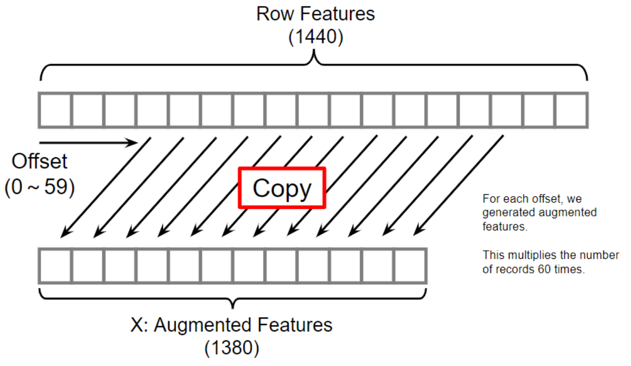


FIGURE 7: Framework of Data Augmentation

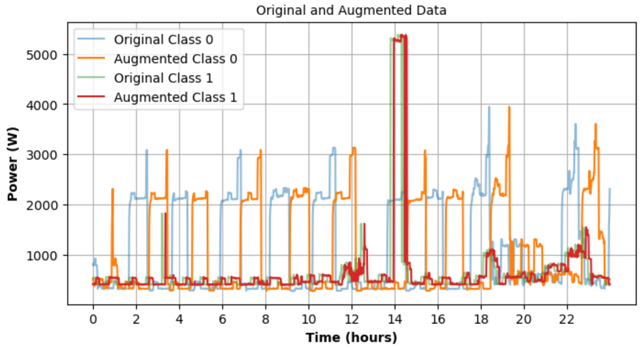


FIGURE 8: Electricity power consumption pattern of Home A for original class and augmented data class for a day

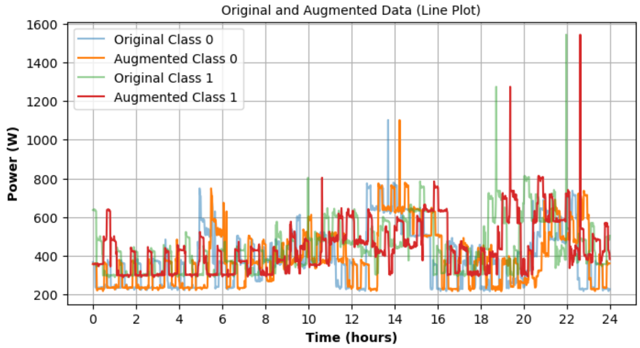


FIGURE 9: Electricity power consumption pattern of all homes for original class and augmented data class for a day

2) Data Augmentation

To increase the number of Benign records, we have applied data augmentation as depicted in Figure 7.

We introduced time offsets ranging from 0 to 59 minutes to create variations of our input features (X) and corresponding labels (Y) for a total of 60 times. Our original dataset consists of pairs of features (X) and labels (Y). Each feature vector X has 1380 elements, representing the power (W) of each minute of the day for 23 hours (1380 minutes). We created a loop that iterates through each sample in our original dataset. For each sample in the loop, we generated a random time offset between 0 and 59 minutes. This offset determined how much shifting in the original feature vector in terms of time. We shifted the 1380-minute feature vector (X) by the generated time offset. This means that we moved the data points in the time series by the specified number of minutes using circular shifting techniques [32] to handle wrap-around effects as necessary. Since we created variations of the same sample, we kept the label (Y) unchanged for each augmented sample as shown in Figure 6. The original sample is labeled as "benign" (0) or "attack" (1), we maintained the same label for all 60 augmented versions of that sample. We repeated these steps for all samples in our original dataset, creating 60 augmented versions of each sample as shown in Figure 8 for the power consumption pattern of a single home, for example Home A, and Figure 9 for power consumption patterns of all homes: Home A, Home B, and Home C respectively.

In our daily life, the operation of home appliances may be shifted for about one hour. Based on this idea, we have shifted the original data over the time axis up to 60 minutes and extracted them also as a benign record.

TABLE 3: Distribution of simulated and real binary dataset

Dataset	Home	Benign samples	Attack samples
SYN Train	A	35040	33818
	B	35040	33242
	C	35040	31533
UN Train	A	35040	0
	B	35040	0
	C	35040	0
Sim Test	A	146	144
	B	146	142
	C	146	133
Real Test	A	8760	8760
	B	8760	8760
	C	8760	8760

Note: SYN = Synthetic, UN = Unsupervised

C. UTOKYO DATA - REAL ATTACK DATA

To test the performance of ETD in both synthetic and unsupervised approaches, we consider including the consumption pattern of real rooms of the University of Tokyo as attack data for model tests. We measured power consumption at the power distribution boards of the I-REF building (6th-floor building) from 2012 with a sampling frequency of 1 minute. We picked up the daily consumption patterns in which the attack impact (AI) corresponds to the attack of the simulation test dataset.

In general, if the AI of the energy theft is large, it could be easily detected with higher accuracy. If the AI of the energy theft is small, it could be negligible – i.e. it does not need to be detected. So, we picked up well-balanced power consumption for evaluation from the real building facilities.

D. DATA PROFILES

Figure 5 illustrates the data samples using the Uniform Manifold Approximation and Projection (UMAP) technique for homes A, B, and C [31]. Each table in this figure corresponds to data from a single day. Notably, different homes exhibit distinct data characteristics. For instance, in Home A, benign data is dispersed and overlaps with various attack samples. Conversely, Home C displays more discernible clusters that may facilitate classification.

However, Table 3 introduces additional data details and parameters for ETA detection. We expanded the number of benign records, resulting in 35,040 benign training records for each home, sourced from monitoring data. Simulated attacks were introduced based on binary classification, as elaborated in Sections V and VI. The table reveals that we have "0" instances for unsupervised training data across all homes. The anomaly detectors were trained solely on benign samples and then subjected to testing on datasets containing both benign and simulated attack samples. Their performance was further evaluated using real attack samples.

E. ATTACK IMPACT

We define attack impact (AI) as a score to measure electricity theft. This attack impacts is the amount of stolen energy in the bill.

Let w_i be the weighted price of the power of x_i on the day at each index i. Let us denote the vector of w_i by $\mathbf{w}$.

The attack impact of electricity theft of the day is :

$$I = \mathbf{w} \cdot \mathbf{x_A} = \sum_i w_i x_{Ai} \quad (2)$$

Here, x_{Ai} represents the power stolen at index i of $\mathbf{x_A}$.

For calculating AI on our dataset, we have assumed 0.20 USD/kWh constantly for the unit price. The daily bill mounted by the attacker is around 1 USD on average (see, Table 2). It will be about 30 USD monthly and 360 USD in a year. If the base power consumption of the home is larger, such as in the case of an office, shop, restaurant, or factory, the attacker will be able to steal much more power.

V. ETD EXPERIMENTS

For our investigation, we deployed Python Jupyter Notebook 3.9.16 and Keras [33] with TensorFlow as the backend. We conducted our experiment with an Intel Core i9 CPU 2.50GHz, 16GB RAM. Windows 11 (64-bit), and NVIDIA GeForce RTX 3070 Ti Laptop GPU. Table 7 provides experi-

TABLE 4: The Configuration of Synthetic Attack Data For Supervised Learning in the experiment

Attack Class	Configuration
Baseload	Theft = 100W
Weakload	Threshold=500W Theft=500W
Peakhour	Threshold=1500W Theft=2000W
Midnight	Starting from 1 AM - 2 AM (at random) Duration = 120 minutes Theft=1000W
Evil-Twin	One Day was randomly chosen in the dataset.

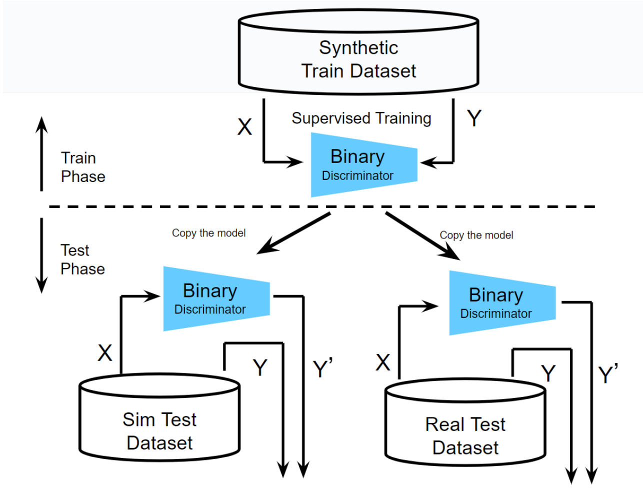


FIGURE 10: Flow of evaluation with Supervised Binary Descriminator

TABLE 5: ETD Models Evaluation with AUC and Accuracy Scores

Home	DataSet	Model	AUC	ACC
A	Sim	SYN-XGB	**98.76%**	95.21%
		SYN-RF	96.47%	91.97%
		SYN-MLP	96.56%	**97.33%**
		UN-MLP-AE	74.61%	58.28%
		UN-1D-CONV-AE	67.31%	54.83%
		UN-IF	63.55%	63.79%
	Real	SYN-XGB	**98.91%**	**95.55%**
		SYN-RF	97.02%	92.23%
		SYN-MLP	96.68%	92.43%
		UN-MLP-AE	58.96%	51.26%
		UN-1D-CONV-AE	61.12%	50.51%
		UN-IF	53.66%	53.66%
B	Sim	SYN-XGB	**98.78%**	**96.15%**
		SYN-RF	97.56%	93.59%
		SYN-MLP	97.24%	95.14%
		UN-MLP-AE	78.90%	64.58%
		UN-1D-CONV-AE	77.04%	65.63%
		UN-IF	61.64%	62.15%
	Real	SYN-XGB	**98.78%**	**96.24%**
		SYN-RF	97.64%	93.27%
		SYN-MLP	97.95%	94.41%
		UN-MLP-AE	60.61%	52.58%
		UN-1D-CONV-AE	71.77%	53.58%
		UN-IF	55.01%	55.01%
C	Sim	SYN-XGB	**98.53%**	**95.26%**
		SYN-RF	96.25%	92.69%
		SYN-MLP	95.37%	91.68%
		UN-MLP-AE	75.04%	55.20%
		UN-1D-CONV-AE	64.95%	60.22%
		UN-IF	67.66%	68.82%
	Real	SYN-XGB	**98.53%**	**95.70%**
		SYN-RF	96.54%	92.91%
		SYN-MLP	96.49%	92.93%
		UN-MLP-AE	65.64%	50.26%
		UN-1D-CONV-AE	54.95%	51.10%
		UN-IF	74.35%	74.35%

TABLE 6: Flow of the evaluation with Synthetic Binary Discriminator

Home	DataSet	Model	Acc	Recall	F1-score	Prec	AUC
A	Sim	XGB	95.21%	0.9169	0.9492	0.9839	98.76%
		RF	91.97%	0.8452	0.9113	0.9886	96.47%
		MLP	97.33%	0.9633	0.9723	0.9816	96.56%
	Real	XGB	95.55%	0.9209	0.9530	0.9875	98.91%
		RF	92.23%	0.8505	0.9146	0.9894	97.02%
		MLP	92.43%	0.9048	0.9212	0.9383	96.68%
B	Sim	XGB	96.15%	0.9281	0.9590	0.9921	98.78%
		RF	93.59%	0.8718	0.9296	0.9957	97.56%
		MLP	95.14%	0.9237	0.9486	0.8718	97.24%
	Real	XGB	96.24%	0.9293	0.9599	0.9927	98.78%
		RF	93.27%	0.8662	0.9259	0.9942	97.64%
		MLP	94.41%	0.9250	0.9413	0.9583	97.95%
C	Sim	XGB	95.26%	0.9116	0.9480	0.9837	98.53%
		RF	92.69%	0.8539	0.9168	0.9906	96.25%
		MLP	91.68%	0.8791	0.9168	0.9406	95.37%
	Real	XGB	95.70%	0.9209	0.9540	0.9894	98.53%
		RF	92.91%	0.8612	0.9216	0.9911	96.54%
		MLP	92.93%	0.9026	0.9251	0.9487	96.49%

TABLE 7: Parameter settings

Model	Parameters	Values
XGB	learning rate	0.01, 0.1
	n_estimators	50, 100
RF	n_estimators	50, 100
	max_depth	5, 10
MLP	hidden_layer_sizes	50,100
	activation	tanh, relu
	batch_size	50, 100
	learning_rate_init	0.001, 0.01
MLP_AE	learning rate	0.001
	hidden_layer_sizes	32
	batch_size	32
	epoch	100
	optimizer	adam
	Validation_split	0.1
1D-CONV_AE	learning rate	0.001
	latent_dim	64
	batch_size	32
	epoch	100
	optimizer	adam
	Validation_split	0.1
IF	n_estimators	100
	random state	42
	contamination	0.05

mental settings regarding attack detection-based ETD binary classification models. For Extreme Gradient Boosting(XGB), we have set up, (1) the learning rate which shrinks the contribution of each tree to values between 0.01 and 0.1, (2) the number of boosting stages to perform n_estimator to values between 50 and 100. For the Random Forest (RF) estimator, we used hyperparameter values with n_estimator between 50 and 100. For Multilayer Perceptron (MLP) 720 features for the first hidden layer and 200 features for the second hidden layer. We have run max_iteration 500 for the training and tanh and relu are used as an activation function output, batch_size of between 50 and 100, and learning rate between 0.001 and 0.01 respectively.

A. ELECTRICITY THEFT DETECTION BASED BINARY DISCRIMINATOR

Figure 10 depicts the architecture of the ETD approach using a supervised learning algorithm. After the training, our classifier can detect electricity attack (Y') and the consumption benign class elements(X).

During the Training Phase, we employed a binary discriminator model with input features denoted as X and target values referred to as Y to adeptly differentiate between instances of electricity consumption patterns, classifying them as either normal (benign) or indicative of electricity theft (attack). Subsequently, in the Testing Phase, our trained model underwent further training using synthetic benign data (X) and simulated attack data (Y) to assess its ability to classify anomalies (Y'), representing unusual electricity consumption patterns. In our evaluation experiment, we put our model to the test with real attack data (X), deploying the binary discriminator to classify consumption patterns and identify anomalies (Y').

For binary classification, we trained our model with a variety of classifiers, including Extreme Gradient Boosting (XGB), Random Forest (RF), and Multi-Layer Perceptron (MLP). Additionally, we explored neural networks as part of our classification approach.

We perform our experiment by splitting the dataset into training and test sets. Trained the model on the training data, using the features as inputs and the corresponding labels (0 or 1) as target outputs. We evaluated the trained model's performance on a separate real test dataset as shown in Table 3, which the model hasn't seen before. Table 6 shows the results of our models. We use the Area Under the Receiver Operating Characteristic Curve (AUC) score as the main evaluation metric. Others include accuracy, precision, recall, F1-score, and the receiver operating characteristic (ROC) curve.

We evaluated the final model's performance by comparing the results on a simulated test and real test dataset(from the UTokyo building), these data values consist of instances not seen during training or evaluation as shown in Figure 10. Unsupervised Learning can detect attacks, however, as we compared that with supervised learning, there are clear differences.

Exploring additional features, such as historical consumption data or external factors like weather, could contribute to better detection accuracy. By employing this approach, the supervised learning model learns to differentiate between normal electricity usage and attack instances, through consumption appliance patterns and helps smart homeowners identify cases of electricity theft and take appropriate action.

B. ELECTRICITY THEFT DETECTION BASED ON UNSUPERVISED LEARNING

Our experimental settings regarding attack detection based on unsupervised binary classification models [34]: Multi-Layer Perceptron Autoencoder (MLP_AE): A fixed learning rate of 0.001 was employed. The autoencoder had a single hidden layer consisting of 32 neurons. A batch size of 32 was used for training. The model was trained for 100 epochs and We utilized the Adam optimizer. A validation split of 10% was employed during training.

1D Convolutional Autoencoder (1D-CONV_AE): A fixed learning rate of 0.001 was employed. The latent space dimension was set to 64. A batch size of 32 was used during training. The model was trained for 100 epochs. Adam optimizer was utilized and a validation split of 10% was employed during training. We used 100 trees in the IF. The random state was set to 42 for reproducibility. The contamination parameter was set to 0.05, representing the assumed proportion of outliers in the dataset.

We selected these hyperparameters to conduct a rigorous evaluation of each model's performance while taking into account various aspects of their architectures and configurations. The choice of hyperparameters was guided by a combination of domain knowledge and empirical experimentation to ensure robust and reproducible results.

The settings presented in Table 7 served as the foundation for our experiments, enabling us to investigate the ability of each model to accurately detect electricity theft. Figure 11 describes the flow of evaluation with unsupervised learning algorithms. Unlike the supervised binary discriminator, we first normalized the data with MinMaxScaler and trained the autoencoder with historical electricity consumption data (benign only) (X) in the Train Phase. We trained our model with Median Absolute Deviation (MAD) to determine the threshold. MAD has an adjustable value to give us the best threshold that will be deployed for the Test Phase. In the test phase, the trained model is replicated with the trained threshold, only simulated test data (X) and attack data (Y) were retrained and evaluated by AUC score, accuracy (Acc), precision (Prec), recall, and F1-score to confirm their effectiveness for anomaly detection. We repeated the same process for the real attack dataset to validate our synthetic simulated approach.

Equations (3), (4), (5), and (6) show each metric, where TP, TN, FP, and FN are true positive, true negative, false positive, and false negatives respectively. TP refers to a sample that is malicious and detected as malicious. TN indicates a benign sample that is detected as benign. FP means that the sample is benign but detected as malicious. FN represents the malicious sample that is detected as benign [35].

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{3}$$

$$Precision = \frac{TP}{TP + FP} \tag{4}$$

$$Recall = \frac{TP}{TP + FN} \tag{5}$$

$$F1 - Score = \frac{2 * Precision * Recall}{Precision + Recall} \tag{6}$$

For our electricity consumption data from various homes and datasets, Equation 7 facilitates the scaling of input features. This scaling ensures that the features, such as power usage or voltage, have uniform scales across different homes and datasets thereby, the original feature values (X) are transformed into scaled values X_{scaled} within the [0, 1] range. This standardized scaling process is essential for the autoencoder

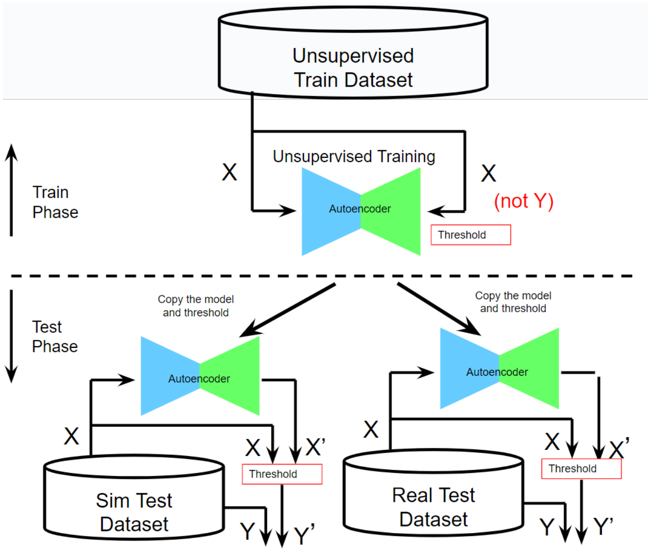


FIGURE 11: Flow of evaluation with Unsupervised Autoencoder

to accurately learn and detect anomalies in electricity consumption patterns. During training, the autoencoder leverages these scaled features to reconstruct benign data, and during testing, a threshold is determined using the statistical method Median Absolute Deviation (MAD), Equations 8 and 9, on reconstruction errors to identify anomalies. This feature scaling contributes to the robustness and accuracy of anomaly detection in the context of protecting homeowners from energy theft by identifying unusual electricity consumption patterns, as indicated in our experimental data from different homes in Table 3.

The Min-Max Scaling formula is given by:

$$X_{\text{scaled}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \tag{7}$$

where:

$$\begin{aligned} X_{\text{scaled}} &: \text{Scaled value of } X \\ X &: \text{Original value of the feature} \\ X_{\min} &: \text{Minimum value of the feature} \\ X_{\max} &: \text{Maximum value of the feature} \end{aligned}$$

1) ETA BASED AUTOENCODER DETECTION ALGORITHM

Electricity Theft Attacker (ETA) can be detected by abnormal electrical consumption features that differ from normal users. Take the user of electricity theft and the normal user as two labels and the detection of electricity theft can be seen as a binary classification problem. For this type of problem, Multi-layer Perceptron Autoencoder (MLP-AE), One Dimensional Convolutionary Autoencoder (1D-CONV-AE), and Isolation Forest (IF) outliers detectors have demonstrated some detection practicality but could not perform well enough compared

TABLE 8: Legacy unsupervised AE Threshold values for Anomaly detection

Home	Dataset	Model	Threshold
A	Sim & Real	UN-MLP-AE	0.017565
		UN-1D-CONV-AE	0.000020
B		UN-MLP-AE	0.018085
		UN-1D-CONV-AE	0.000055
C		UN-MLP-AE	0.019676
		UN-1D-CONV-AE	0.000055

to XGB, RF and MLP classifiers with reference AUC metric scores.

In our model, we deployed a reconstruction error threshold to classify data points as normal (benign) or anomalous (attack) and also metrics like area under the receiver operating characteristic curve (AUC-ROC), F1-score, precision, and recall, which are often more informative in such cases. High accuracy can be achieved by simply classifying everything as benign, which does not help detect attacks.

2) Threshold Determination:

In our experiment, we computed the threshold by training the model with benign data only as mentioned in the previous section. For the Autoencoder, generating ROC curves and establishing threshold values, we employ the robust z-score 0.75th quartile of the reconstruction error distribution of the validation set to select as our error threshold. In Figure 13 we plotted the ROC curves for the MLP-AE, 1D-CONV-AE, and IF for both simulated and real attack scenarios in Home A. MLP-AE detector outperform both 1D-CONV-AE and IF with AUC score of 0.76 for simulated and 0.59 for real attacks while 1D-CONV-AE has the ROC value 0.67 and 0.61; IF has AUC values 0.64 and 0.54 respectively. MAD Was employed to determine the ideal threshold values for the anomaly detectors to distinguish between benign and malicious samples.

Our autoencoder model was trained only with benign sample (X) as shown in Figure 11 to determine the threshold values for each of the model MLP-AE and 1D-CONV-AE for each of the home, both for the simulated(sim) attack and real home attack. The optimal threshold values for ETD are depicted in Table 8 for all the homes.

Figures 12 and 13 further generates ROC curves for the remaining supervised benchmark detectors to facilitate comparative analysis. In the Z-score method, the median is a robust statistic, meaning it will not be greatly affected by outliers. This is called the Robust Z-score method, and instead of using standard deviation, it uses the MAD (Median Absolute Deviation) [36].

The mathematical representation of the Modified Z-score is:

$$\text{MAD} = \text{median}\left(|x_i - \tilde{x}|\right) \tag{8}$$

$$\text{Modified Z-score} = 0.6745 \times \frac{x_i - \tilde{x}}{\text{MAD}} \tag{9}$$

0.6745 is the 0.75th quartile of the standard normal distribution, to which the MAD converges.

- $\tilde{x}$ which is just the median of the sample
- MAD, is calculated by taking the absolute difference between each point and the median, and then calculating the median of those differences.

In many real-world scenarios, attack patterns are varied and constantly evolving. Rare attack patterns can be vastly exceeded by benign data. As a result, the autoencoder may not have sufficient examples of these rare attacks to learn effective representations, making it difficult to detect new attacks.

VI. EXPERIMENTS RESULTS AND PERFORMANCE EVALUATION

We present a comprehensive performance evaluation of various machine learning models for ETD, utilizing both real-world and synthetic attack datasets. We primarily focus on the AUC metric from Table 5 as the primary evaluation criterion and complement it with additional metrics from Table 6, including F1-score, accuracy, precision, and recall.

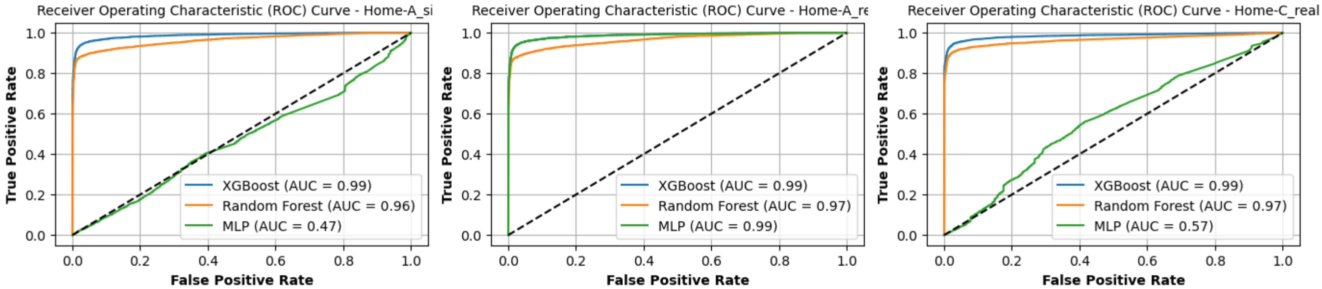
1) Model Performance on Synthetic Attack Data

For Home A, when evaluated on simulated synthetic attack data, for example, the SYN-XGB model stands out with an impressive AUC score of 98.76%. It achieves a high F1-score of 94.92%, indicating robust performance in capturing fraudulent electricity consumption patterns as illustrated in Table 6. SYN-RF and SYN-MLP also exhibit strong AUC scores of 96.47% and 96.56% respectively, indicating their efficacy in identifying anomalies. It is noteworthy that SYN-MLP has a slightly higher AUC score metric performance detection rate of 0.06% than SYN-RF and also has a higher F1-score of 97.23% which is 2.31% and 6.10% performance rate better than SYN-XGB and SYN-RF respectively in Home A simulated attack only, suggesting a trade-off between precision and recall.

In contrast, the legacy unsupervised models(LUM), UN-MLP-AE, UN-1D-CONV-AE, and UN-IF, struggle to match the performance of supervised models on synthetic data. UN-MLP-AE achieves an aggregated AUC score of 76.18% and 61.74%, while UN-1D-CONV-AE (69.77% and 62.61%), and UN-IF (64.28%), for simulated and real attacks for all homes, lag further behind, respectively as depicted in Table 9.

2) Model Performance on Real Attack Data

When assessing the same models on real attack data, for example in Home A, SYN-XGB continues to excel with an AUC score of 98.91%, reaffirming its capability to detect electricity theft accurately. Similarly, SYN-RF and SYN-MLP maintain strong AUC scores of 97.02% and 96.68%, respectively. These models also exhibit competitive F1-scores, indicating their reliability in real-world scenarios.

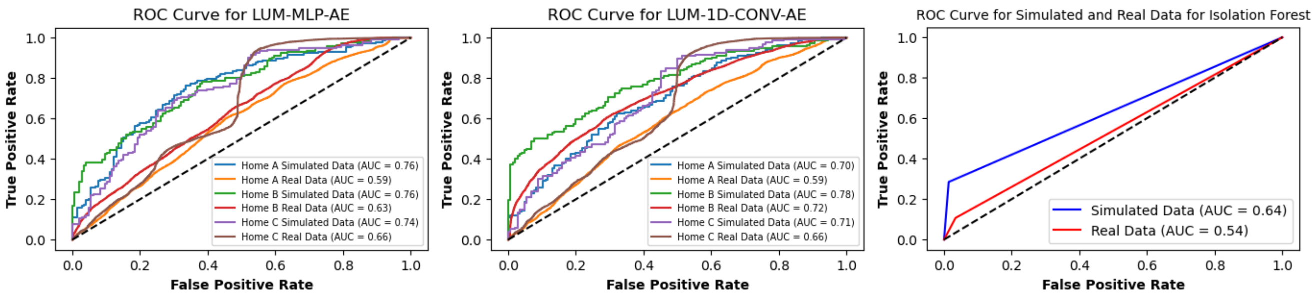


(a) ROC curves for three models based simulated attack in Home A

(b) ROC curves for three models based on real attack in Home A

(c) ROC curves for three models based on a real attack in Home C

FIGURE 12: Sampled ROC curves comparison for performance evaluation of our proposed synthetic ETD of some selected homes for simulated and real attack.



(a) MLP-AE: sim vs. real attack

(b) 1D-CONV-AE: sim vs. real attack

(c) IF: sim vs. real attack

FIGURE 13: Performance comparison of parameter tuning for legacy unsupervised models.

In contrast, the LUM's performance drops significantly when faced with a real attack scenario. UN-MLP-AE, UN-1D-CONV-AE, and UN-IF struggle to achieve AUC scores above 60%, signaling their limitations in detecting electricity theft in practical settings.

3) Performance by Attack Class

Figure 14 shows the confusion matrices of the trained models for Home A simulated and real attacks class. The colors of this figure are:

- Red if the overall accuracy is more than 95%.
- Blue if the overall accuracy is more than 92%
- Green if the overall accuracy is more than 91% respectively.
- Gray if the overall accuracy is between 60% and 80%.

The numbers (i.e., No. #) indicate the rank, which is corresponded to Table 6.

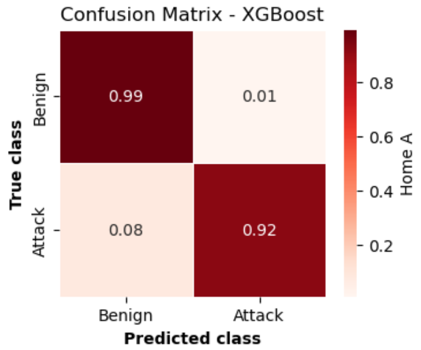
We can observe that the anomalies were well classified in all the synthetic classifiers. Especially, with XGB, RF, and MLP – i.e., the best three algorithms, have achieved average performance of between 92.75% for RF simulated attack and 95.92% for XGB real attack detection accuracy for all homes.

We compared LUM anomaly detectors, as shown in Figure 9, UN-IF performs better with an average accuracy score of 64.92% and 61.01% for the simulated attacks and real attacks. UN-1D-CONV-AE followed with average accuracy scores of 60.23% and 51.70% while UN-MLP-AE had 59.35% and 51.37% for the same attack scenario for all homes respectively.

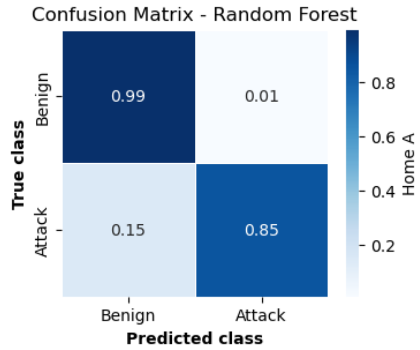
These false positives are very critical in our real attack operational scenarios, hence we will investigate further in our next experiment. XGB, RF, and MLP also have false positives in attack detection. However, the ratio has decreased to less than 1%. They have false negatives in attack detection with 8-15%, which can be eventually detected if they continuously perform the attacks every day onwards. On the contrary, the unsupervised UN-IF, UN-1D-CONV-AE, and UN-MLP-AE have high false negatives in attack detection with 46-78% which makes these models difficult for overall accurate detection compared to the synthetic models.

4) Comparison Across Homes

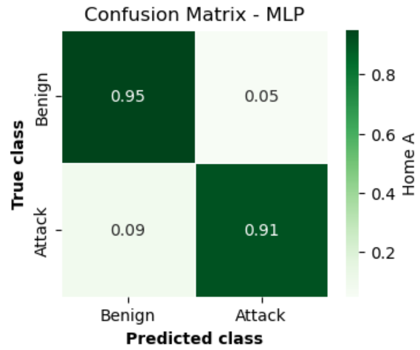
Across different households (A, B, and C), the supervised models consistently outperform the unsupervised models on both synthetic and real attack data. SYN-XGB emerges as the top performer with consistently high AUC scores, making it a robust choice for electricity theft detection.



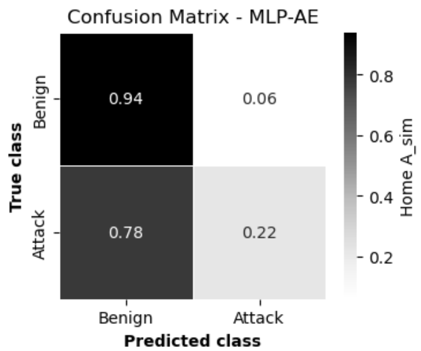
(No.1) XG Boosting (XGB)



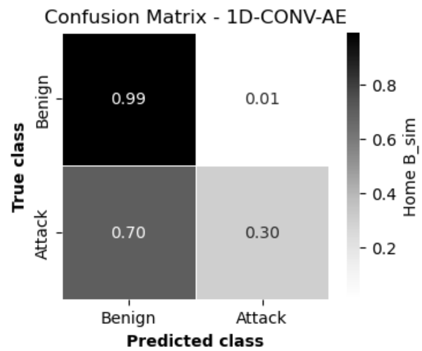
(No.2) Random Forest (RF)



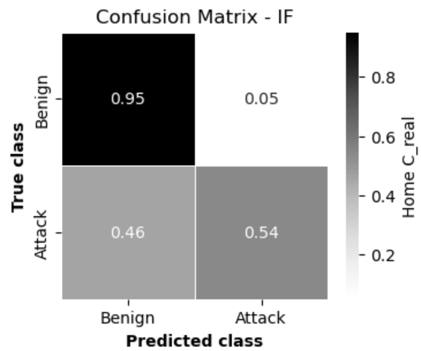
(No.3) Multilayer Perceptron (MLP)



(No.4) Multilayer Perceptron Autoencoder



(No.5) 1D-Convolutional Autoencoder



(No.6) Isolation Forest Outliers (IF)

FIGURE 14: Confusion matrices of the trained models. The number (No. #) indicates the rank of the overall accuracy (table 6). The colors indicate the group of accuracy. The attacks class were relatively easily detected by all the synthetic models, especially achieving 99% accuracy with XGB and RF. The false positive rates of XGB, and RF were about 1% which is much better than other models.

TABLE 9: Aggregate AUC and ACC values of our models for all Homes

Home	DataSet	Model	AUC	ACC
A+B+C	Sim_av	SYN-XGB	**98.69%**	**95.54%**
		SYN-RF	96.76%	92.75%
		SYN-MLP	96.32%	94.72%
		UN-MLP-AE	76.18%	59.35%
		UN-1D-CONV-AE	69.77%	60.23%
		UN-IF	64.28%	64.92%
	Real_av	SYN-XGB	**98.74%**	**95.83%**
		SYN-RF	97.07%	92.80%
		SYN-MLP	97.04%	93.26%
		UN-MLP-AE	61.74%	51.37%
		UN-1D-CONV-AE	62.61%	51.70%
		UN-IF	64.28%	61.01%

VII. DISCUSSION AND FUTURE WORKS

In this study, we conducted a comprehensive evaluation of machine learning models for electricity theft detection using both real-world and synthetic attack datasets. Our findings shed light on the effectiveness of different models in identifying abnormal consumption patterns indicative of electricity theft. We also considered the implications of these findings for improving the security of home appliance-based electricity consumption monitoring systems.

We contribute to this body of research by systematically evaluating the performance of various ML models, including supervised and unsupervised approaches, on both real and simulated attack data. Our focus on home appliance consumption patterns allows us to gain insights into the strengths and weaknesses of these models in detecting electricity theft scenarios related to specific appliances.

The detection of covert theft schemes that closely mimic legitimate consumption patterns remains a difficult problem. Additionally, scaling these detection systems to accommodate large-scale smart grids and diverse consumer behaviors presents ongoing challenges.

One prominent limitation lies in the model's reliance on synthetic attack data. While this approach offers controlled experimentation, it raises questions about the model's ability to faithfully replicate the diversity and dynamism of real-world attack scenarios. The debate surrounding the extent to which synthetic data can accurately represent these intricate scenarios remains an important point of consideration.

Furthermore, although our model has undergone evaluation with real-world attack data, the volume of available data is still limited. To enhance the model's real-world applicability, a more extensive and diverse set of real attack scenarios should be explored. This would provide a more comprehensive understanding of the model's capabilities in diverse contexts.

The model's current performance discussions are rooted in the specific homes included in our study (Home A, Home B, and Home C). However, smart homes come in various configurations, and the generalizability of our approach to a broader range of smart homes needs further investigation.

As data privacy concerns continue to grow, ensuring that mechanisms for data collection, especially real-world attack data, comply with privacy standards is imperative. These can be addressed, by developing techniques that can detect theft without compromising customer privacy, such as federated learning or secure multiparty computation in smart home energy consumption monitoring [37], [38].

Collaboration with regulatory authorities and compliance with emerging regulations in the smart grid and cybersecurity domains are essential for the model's deployment.

In practical applications, effectively communicating detection results and integrating the system with user-friendly interfaces for homeowners and utility companies is vital. This involves exploring the ease of use and interpretability of the model's output to ensure its practical utility.

In future works, we plan to address these limitations knowing well that embarking on further research in these areas will contribute to the ongoing development of robust and effective ETD systems for smart homes. As the field continues to evolve, these challenges provide exciting avenues for future exploration and innovation in the quest for more secure and reliable smart grids.

VIII. CONCLUSION

Our study highlights the significance of machine learning models in the detection of electricity theft, particularly concerning home appliance consumption patterns. Supervised learning models, especially XGB and RF models, exhibited robust performance in various scenarios, making them suitable choices for electricity theft detection tasks. However, these models demonstrated some sensitivity to the complexities of real-world electricity consumption patterns. Unsupervised learning models, while effective in simulated scenarios, struggled to adapt to the intricacies of real attack data, indicating the need for further research to enhance their performance in real-world settings. Overall, our findings provide valuable insights into the application of machine learning in energy security and emphasize the importance of considering home appliance consumption patterns in the detection of electricity theft, benefiting both consumers and energy providers in achieving more efficient and secure energy usage.

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